

ANSWERS

1.

- (a) sum of kinetic and potential energies of molecules / particles / atomsM1
 random (distribution) A1

- (b) $+\Delta U$: increase in internal energy B1
 $+q$: heating of / heat supplied to system B1
 $+w$: work done on system B1

- (c) (i) work done = $p\Delta V$ C1
 $= 1.0 \times 10^5 \times (2.1 - 1.8) \times 10^{-3}$
 $= 30 \text{ J}$ M1
 $w = 30 \text{ J}, q = 0$ so $\Delta U = 30 \text{ J}$ A1

2.

- (a) (i) no forces (of attraction or repulsion) between atoms / molecules / particles B1

- (ii) sum of kinetic and potential energy of atoms / molecules M1
 due to random motion A1

- (iii) (random) kinetic energy increases with temperature M1
 no potential energy
 (so increase in temperature increases internal energy) A1

- (b) (i) zero A1

- (ii) work done = $p\Delta V$ C1
 $= 4.0 \times 10^5 \times 6 \times 10^{-4}$
 $= 240 \text{ J}$ (ignore any sign) A1

(iii)

change	work done / J	heating / J	increase in internal energy / J
P $\rightarrow$ Q	+240	-600	-360
Q $\rightarrow$ R	0	+720	+720
R $\rightarrow$ P	-840	+480	-360

(correct signs essential)
 (each horizontal line correct, 1 mark – max 3)

B3

3.	
(a) $+\Delta U$: increase in internal energy	B1
$+q$: thermal energy / heat supplied to the system	B1
$+w$: work done on the system	B1
(b) (i) (thermal) energy required to change the state of a substance per unit mass without any change of temperature	M1 A1 A1
(ii) when evaporating greater change in separation of atoms/molecules greater change in volume identifies each difference correctly with ΔU and w	M1 M1 A1
4.	
(a) sum of potential energy and kinetic energy of atoms/molecules/particles reference to random (distribution)	M1 A1
(b) (i) as lattice structure is 'broken'/bonds broken/forces between molecules reduced (not molecules separate) no change in kinetic energy, potential energy increases internal energy increases	B1 M1 A1
(ii) <i>either</i> molecules/atoms/particles move faster/ $\langle c^2 \rangle$ is increasing <i>or</i> kinetic energy increases with temperature (increases) no change in potential energy, kinetic energy increases internal energy increases	B1 M1 A1
5.	
(a) (i) sum of potential energy and kinetic energy of atoms / molecules / particles reference to random	M1 A1
(ii) no intermolecular forces no potential energy internal energy is kinetic energy (of random motion) of molecules <i>(reference to random motion here then allow back credit to (i) if M1 scored)</i>	B1 B1 B1
(b) kinetic energy $\propto$ thermodynamic temperature <i>either</i> temperature in Celsius, not kelvin so incorrect <i>or</i> temperature in kelvin is not doubled	B1 B1

6.

- (a) (i) sum of kinetic and potential energies of the molecules
reference to random distribution M1
A1
- (ii) for ideal gas, no intermolecular forces M1
so no potential energy (only kinetic) A1
- (b) (i) *either* change in kinetic energy $= \frac{3}{2} \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23} \times 180$ C1
 $= 2240 \text{ J}$ A1
or $R = kN_A$
energy $= \frac{3}{2} \times 1.0 \times 8.31 \times 180$ (C1)
 $= 2240 \text{ J}$ (A1)
- (ii) increase in internal energy = heat supplied + work done on system B1
 $2240 = \text{energy supplied} - 1500$ C1
energy supplied = 3740 J A1

7.

- (a) *either* change in volume $= (1.69 - 1.00 \times 10^{-3})$
or liquid volume $\ll$ volume of vapour M1
work done $= 1.01 \times 10^5 \times 1.69 = 1.71 \times 10^5 \text{ (J)}$ A1
- (b) (i) 1. heating of system/thermal energy supplied to the system B1
2. work done on the system B1
- (ii) $\Delta U = (2.26 \times 10^6) - (1.71 \times 10^5)$ C1
 $= 2.09 \times 10^6 \text{ J}$ (3 s.f. needed) A1

8.

- (a) (sum of) potential energy and kinetic energy of molecules/atoms/particles M1
mention of random motion/distribution A1
- (b) (i) $pV = nRT$
either at A, $1.2 \times 10^5 \times 4.0 \times 10^{-3} = n \times 8.31 \times 290$
or at B, $3.6 \times 10^5 \times 4.0 \times 10^{-3} = n \times 8.31 \times 870$ C1
 $n = 0.20 \text{ mol}$ A1
- (ii) $1.2 \times 10^5 \times 7.75 \times 10^{-3} = 0.20 \times 8.31 \times T$ *or* $T = (7.75/4.0) \times 290$ C1
 $T = 560 \text{ K}$ A1
(Allow tolerance from graph: $7.7\text{--}7.8 \times 10^{-3} \text{ m}^3$)
- (c) temperature changes/decreases so internal energy changes/decreases B1
volume changes (at constant pressure) so work is done B1

9.

(a) heat/thermal energy gained by system *or* energy transferred to system by heating plus work done on the system *or* minus work done by the system B1
B1

(b) (i) *either* volume decreases so work done on the system M1
or small volume change so work done on system negligible M1
(thermal) energy absorbed to break lattice structure A1
internal energy increases

(ii) gas expands so work done by gas (against atmosphere) M1
no time for thermal energy to enter or leave the gas M1
internal energy decreases A1

10.

(a) (i) $p \propto T$ *or* $pV/T = \text{constant}$ *or* $pV = nRT$ C1

$T (= 5 \times 300 =) 1500 \text{ K}$ A1

(ii) $pV = nRT$

$$1.0 \times 10^5 \times 4.0 \times 10^{-4} = n \times 8.31 \times 300$$

or

$$5.0 \times 10^5 \times 4.0 \times 10^{-4} = n \times 8.31 \times 1500$$
 C1

$$n = 0.016 \text{ mol}$$
 A1

(b) (i) 1. heating/thermal energy supplied B1

2. work done on/to system B1

(ii) 1. 240 J A1

2. same value as given in 1. (= 240 J) **and** zero given for 3. A1

3. zero A1

11.

(a) (sum of/total) potential energy and kinetic energy of (all) molecules/particles M1
reference to random (distribution) A1

(b) (i) no heat enters (gas)/leaves (gas)/no heating (of gas) B1

work done by gas (against atmosphere as it expands) M1

internal energy decreases A1

(ii) volume decreases so work done on ice/water
(allow work done negligible because ΔV small)

B1

heating of ice (to break rigid forces/bonds)

M1

internal energy increases

A1

12.

2(a)	+ ΔU increase in internal energy + q heat (energy) transferred <u>to</u> the system / heating of system + w work done <u>on</u> system	B2
2(b)(i)	$W = p\Delta V$ $= 5.2 \times 10^5 \times (5.0 - 1.6) \times 10^{-4} \quad (=177 \text{ J})$	B1
	$\Delta U = q + w$ $= 442 - 177 = 265 \text{ J}$	A1
2(b)(ii)	no (molecular) potential energy	B1
	internal energy decreases so (total molecular) kinetic energy decreases	B1
	(mean molecular) kinetic energy decreases so temperature decreases	B1
2(b)(iii)	$\Delta U + 265 - 313 = 0$ $\Delta U = 48 \text{ J}$	A1
2(b)(iv)	$pV = NkT$ or $pV = nRT$ <u>and</u> $N = nN_A$	C1
	$5.2 \times 10^5 \times 1.6 \times 10^{-4} = N \times 1.38 \times 10^{-23} \times (273 + 227)$ or $5.2 \times 10^5 \times 1.6 \times 10^{-4} = n \times 8.31 \times (273 + 227)$ and $n = N / 6.02 \times 10^{23}$ $N = 1.2 \times 10^{22}$	A1

13.

3(a)(i)	sum of potential and kinetic energies (of molecules/atoms/particles)	B1
	(energy of) molecules/atoms/particles in random motion	B1
3(a)(ii)	(in ideal gas) no intermolecular forces so no potential energy	B1
	internal energy is (solely) kinetic energy (of particles)	B1
	(mean) kinetic energy (of particles) proportional to (thermodynamic) temperature of gas	B1
3(b)	$pV = NkT$	C1
	$6.4 \times 10^6 \times 1.8 \times 10^4 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 298$	C1
	or	
	$pV = nRT$ <u>and</u> $N = n \times N_A$	(C1)
	$6.4 \times 10^6 \times 1.8 \times 10^4 \times 10^{-6} = n \times 8.31 \times 298$	(C1)
	$n = 46.5 \text{ (mol)}$	
	$N = 46.5 \times 6.02 \times 10^{23}$	
	$N = 2.8 \times 10^{25}$	A1

14.

2(a)	sum of potential and kinetic energies (of molecules/atoms/particles)	B1
	(energy of) molecules/atoms/particles in random motion	B1
2(b)(i)	final temperature = initial temperature	B1
	no change in internal energy	B1
2(b)(ii)	1. work done on gas (P→Q): 0	A1
	increase in internal energy (P→Q): (+)97.0 J	A1
	2. increase in internal energy (Q→R): -42.5 J	A1
	3. increase in internal energy (R→P): -54.5 J	A1
	thermal energy supplied (R→P): -91.5 J	A1

15.

2(a)(i)	1. energy transfer <u>to</u> the system by heating	B1							
	2. (external) work done <u>on</u> the system	B1							
2(a)(ii)	<u>decrease</u> in internal energy	B1							
2(b)(i)	no change (in internal energy)	B1							
	(because) no change in temperature	B1							
2(b)(ii)	work done = $p\Delta V$ $= (-)1.6 \times 10^5 \times (2.4 - 0.87) \times 10^{-3}$	C1							
	= (-)240 J	A1							
2(b)(iii)	first row all correct (0, 480, 480)	A1							
	second row all correct (-1100, 0, -1100)	A1							
	final column of third row calculated correctly from the two values above it, so that the final column adds up to 0	A1							
	second column in final row correct, with correct negative sign and first column in final row calculated correctly so that it adds to the second column to give the third column (fully correct table is:	A1							
	<table border="1"> <tr> <td>0</td><td>480</td><td>480</td></tr> <tr> <td>-1100</td><td>0</td><td>-1100</td></tr> <tr> <td>860</td><td>-240</td><td>620</td></tr> </table>		0	480	480	-1100	0	-1100	860
0	480	480							
-1100	0	-1100							
860	-240	620							

16.

2(a)	total potential energy and kinetic energy (of molecules/atoms)	M1
	reference to <u>random</u> motion of molecules/atoms	A1
2(b)	(in ideal gas,) no intermolecular forces	B1
	no potential energy (so change in kinetic energy is change in internal energy)	B1

2(c)	(random) potential energy of molecules does not change	M1
	(random) kinetic energy of molecules does not change	M1
	so internal energy does not change	A1
	or	
	decrease in total potential energy = gain in total kinetic energy	(M1)
	no external energy supplied	(M1)
	so internal energy does not change	(A1)
	or	
	no compression (of ball) so no work done on the ball	(M1)
	no resistive forces so no heating of the ball	(M1)
	so internal energy does not change	(A1)

17.

3(a)	(little/no volume change so) little/no external work done	B1
	thermal energy supplied to provide latent heat	M1
	internal energy increases	A1
3(b)	(rapid) increase in volume	B1
	gas does work against the atmosphere	M1
	internal energy decreases	A1

18.

2(a)	+q: thermal energy transfer to system	B1																
	+w: work done on system	B1																
2(b)(i)	(W =) $2.6 \times 10^5 \times (3.8 - 2.3) \times 10^{-3} = 390 \text{ J}$	A1																
2(b)(ii)	no (total) change (in internal energy)	B1																
	gas returns to its original temperature	B1																
2(c)	A to B row all correct (1370, - 390, 980)	B1																
	B to C row all correct (0, 550, 550)	B1																
	C to A row: ΔU adds to the other two ΔU values to give zero	B1																
	C to A row: $w = 0$ and q adds to w to give ΔU value	B1																
	complete correct answer:																	
<table><tr><td>change</td><td>q / J</td><td>w / J</td><td>$\Delta U / \text{J}$</td></tr><tr><td>A to B</td><td>(+)1370</td><td>-390</td><td>(+)980</td></tr><tr><td>B to C</td><td>0</td><td>(+)550</td><td>(+)550</td></tr><tr><td>C to A</td><td>-1530</td><td>0</td><td>-1530</td></tr></table>			change	q / J	w / J	$\Delta U / \text{J}$	A to B	(+)1370	-390	(+)980	B to C	0	(+)550	(+)550	C to A	-1530	0	-1530
change	q / J	w / J	$\Delta U / \text{J}$															
A to B	(+)1370	-390	(+)980															
B to C	0	(+)550	(+)550															
C to A	-1530	0	-1530															

19.

3(a)	(thermal) energy per unit mass (to cause temperature change)	B1
	(thermal) energy per unit <u>change</u> in temperature	B1
3(b)(i)	$(T =) pV / Nk$	B1
3(b)(ii)	$(pV =) NkT = \frac{1}{2}Nm\langle c^2 \rangle$ or $pV = NkT$ and $pV = \frac{1}{2}Nm\langle c^2 \rangle$	M1
	leading to $\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ and $\frac{1}{2}m\langle c^2 \rangle = E_K$	A1
3(b)(iii)	internal energy = ΣE_K (of molecules) + ΣE_P (of molecules) or no forces between molecules	B1
	potential energy of molecules is zero	B1
3(c)(i)	increase in internal energy = Q + work done	B1
	constant volume so no work done	B1
3(c)(ii)	$c = Q / Nm\Delta T$	C1
	$= [N \times (3/2)k\Delta T] / (Nm\Delta T) = 3k / 2m$	A1
3(d)	(as it expands) gas does work (against the atmosphere/external pressure)	B1
	for same temperature rise) more (thermal) energy needed, so larger specific heat capacity	B1

20.

2(a)	0	B1
2(b)	$pV = nRT$	C1
	$(n =) 1.5 \times 10^5 \times 4.2 \times 10^{-3} / 8.31 \times 540$ $= 0.14 \text{ mol}$	A1
2(c)	missing pressure 1.5 ($\times 10^5$)	B1
	both missing volumes 1.8 ($\times 10^{-3}$)	B1
2(d)(i)	$(\Delta U:)$ increase in internal energy (of the system)	B1
	$(q:)$ thermal energy supplied to the system	B1
	$(W:)$ work done on system	B1
2(d)(ii)	volume increases and work is done by the gas	B1
	temperature decreases and internal energy decreases	B1

21.

3(a)(i)	a gas that obeys $pV \propto T$	M1											
	where p = pressure, V = volume, T = thermodynamic temperature	A1											
3(a)(ii)	$T = (273 + 17) \text{ K}$	C1											
	$n = pV / RT$	A1											
	$= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273 + 17)]$												
	$= 12 \text{ mol}$												
3(b)(i)	work done = $p\Delta V$	C1											
	$= 1.2 \times 10^5 \times (0.24 - 0.08) = 19200 \text{ J } (= 19.2 \text{ kJ})$	A1											
3(b)(ii)	AB work done correct (19.2)	A1											
	BC work done correct (0)	A1											
	CA increase in internal energy correct (0) and CA thermal energy correct (31.6)	A1											
	AB increase in internal energy calculated correctly from work done – 48.0	A1											
	BC increase in internal energy correctly calculated so the final column adds up to zero and BC thermal energy same as increase in internal energy	A1											
	(Fully correct table: <table><tr><td>AB</td><td>19.2</td><td>–48.0</td><td>–28.8</td></tr><tr><td>BC</td><td>0</td><td>28.8</td><td>28.8</td></tr><tr><td>CA</td><td>–31.6</td><td>31.6</td><td>0</td></tr></table>)	AB	19.2	–48.0	–28.8	BC	0	28.8	28.8	CA	–31.6	31.6	0
AB	19.2	–48.0	–28.8										
BC	0	28.8	28.8										
CA	–31.6	31.6	0										

22.

2(a)	(thermal) energy per unit mass (to cause temperature change)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	work done correct (0)	B1
	increase in internal energy correct (+E)	B1
2(b)(ii)	work done correct ($-W$) and increase in internal energy same as (b)(i)	B1
	thermal energy correct so that it adds to work done to give increase in internal energy	B1
2(c)	more thermal energy needed so specific heat capacity is greater	B1

23.

2(a)	gas for which $pV \propto T$	M1
	where T is thermodynamic temperature	A1
2(b)(i)	evidence of two temperature conversions between $^{\circ}\text{C}$ and K	B1
	two calculations shown, one for each state e.g. $\frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{(273 + 27)} = 0.198 \text{ and } \frac{6.70 \times 10^6 \times 30 \times 10^{-6}}{(273 + 742)} = 0.198$	A1
2(b)(ii)	work is done on the gas	M1
	internal energy increases (so temperature increases)	A1

2(b)(iii)	$pV = NkT$ e.g. $N = \frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$ $= 1.435 \times 10^{22}$ $\Delta E_k = (3/2) k \Delta TN$	C1
	$= (3/2) \times 1.38 \times 10^{23} \times (742 - 27) \times \frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$	C1
	$= 212 \text{ J}$	A1
2(c)	$E = mc\Delta\theta$ and $E = mL$	C1
	$\Delta\theta = (27 + 196)$ or 223	C1
	$E = 0.0240 \times 1.04 \times (27 + 196) + 0.0240 \times 199$ $= 10.3 \text{ kJ}$	A1

24.

4(a)	- particles are in (continuous) random motion - particles have negligible volume (compared with the gas) - negligible forces between particles (except during collisions) (all) collisions (perfectly) elastic - time of collision negligible (in comparison with time between collisions) <i>Any two points, 1 mark each</i>	B2															
4(b)(i)	(general starting equation) $pV = nRT$	C1															
	$T = (2pV / nR)$ where R is the (molar) gas constant	A1															
4(b)(ii)	sketch: straight vertical line XY from $(V, 2p)$ to (V, p)	B1															
	straight horizontal line YZ from (V, p) to $(2V, p)$	B1															
	curve with gradient increasing from Z to X from $(2V, p)$ to $(V, 2p)$	B1															
4(b)(iii)	XY work done on gas correct ($= 0$)	B1															
	ZX increase in internal energy correct ($= 0$)	B1															
	YZ work done on gas correct ($= -pV$)	B1															
	XY increase in internal energy such that the increase in internal energy column adds up to zero	B1															
	all three thermal energies transferred such that $\Delta U = q + w$ in each row	B1															
	(completely correct answer: <table border="1"><thead><tr><th>change</th><th>ΔU</th><th>Q</th><th>w</th></tr></thead><tbody><tr><td>X to Y</td><td>$-U$</td><td>$-U$</td><td>0</td></tr><tr><td>Y to Z</td><td>$[+U]$</td><td>$U + pV$</td><td>$-pV$</td></tr><tr><td>Z to X</td><td>0</td><td>$-W$</td><td>$[+W]$</td></tr></tbody></table>)	change	ΔU	Q	w	X to Y	$-U$	$-U$	0	Y to Z	$[+U]$	$U + pV$	$-pV$	Z to X	0	$-W$	$[+W]$
change	ΔU	Q	w														
X to Y	$-U$	$-U$	0														
Y to Z	$[+U]$	$U + pV$	$-pV$														
Z to X	0	$-W$	$[+W]$														

25.

3(a)	<u>change</u> in internal energy = work done + energy transfer by heating	C1
	<u>increase</u> in internal energy = work done <u>on</u> system + energy transferred <u>to</u> the system by heating	A1
3(b)(i)	AB change in internal energy: decrease	B1
	AB work done on gas: positive	B1
	BC change in internal energy: increase	B1
	BC work done on gas: zero	B1
3(b)(ii)	more work done by gas in CD than is done on gas in AB or (no work done on gas in BC and DA so) (overall) gas does work	B1
	(overall) change in internal energy is zero	B1
	(must be an overall) input of thermal energy	B1

26.

2(a)	(thermal) energy per unit mass (to change temperature)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	work done = $p\Delta V$ $= (2.0 \times 10^5) \times (0.063 - 0.038) = 5000 \text{ J}$	A1
2(b)(ii)	gas is expanding (against external pressure)	B1
	gas does work / work is done by gas, so (work done on gas is) negative	B1
2(b)(iii)	$\Delta U = q + W$	C1
	$7600 = q + (-5000)$	A1
	$q = 12600 \text{ J}$	
2(b)(iv)	specific heat capacity = $q / m\Delta T$	C1
	$= 12600 / (0.35 \times 56)$	
	$= 640 \text{ J kg}^{-1} \text{ K}^{-1}$	A1
2(c)	same gain in internal energy so same temperature rise	B1
	no change in volume so no work done or no work done so less thermal energy needed (for same change in internal energy)	B1
	less thermal energy needed (for same temperature change) so lower specific heat capacity	B1

27.

3(a)	sum of potential energy and kinetic energy (of particles)	B1
	(total) energy of random motion of particles	B1
3(b)(i)	no thermal energy <u>transferred</u>	B1
	work is done on the spring (increasing the potential energy of particles)	M1
	so internal energy increases	A1
3(b)(ii)	thermal energy transferred to water	B1
	work is done by water (expanding against atmosphere as it vaporises)	B1
	more thermal energy transferred than work done so internal energy increases	B1